AI-Driven Academic Risk & Dropout Early-Warning System

Presented by

Altrin Benser A - 962223205010

Jerish Rethna J - 962223205029

John Jins J - 962223205032

Mukhilan S - 962223205040

Guide:

Er. P Agnes Alex Rathy ME., MBA.

Assistant Professor - IT, SXCCE

Abstract

Traditional academic systems mainly identify weak students after semester or examination results and treat student information as static records. This limits continuous monitoring and early intervention for academically struggling students.

Our proposed **Digital Class Twin** continuously collects student attendance, assessment, quiz and behavioral data as events. It maintains an evolving digital representation of each student's academic state over time.

Machine learning techniques analyse these data to identify and classify students based on their academic risk.

Problem Statement

Existing Problems

- Weak students are often identified only after examination/semester results.
- Student data is commonly maintained as static records.
- Static records do not adequately represent changing student performance over time.
- Rule-based approaches such as attendance thresholds cannot capture the complete academic situation.
- Institutions lack an intelligent mechanism for continuous monitoring and early intervention.

Need for the project

To design a system that can continuously monitor student progress, identify declining academic students at an early stage, and provide understandable evidence to teachers and counselors.

Objectives

1. To design a system for continuously collecting student academic and behavioural data.
   - Attendance
   - Assessment
   - Behavioural records
2. To develop a Digital Twin for each student.
   - Current academic state
   - Historical time-series data
   - Synchronization status
3. To develop a machine learning model to predict academic risk as Low, Medium, or High.
4. To design a machine learning-based model for predicting future GPA.
5. To analyze student academic trajectories and classify them as Declining, Stable, or Recovering.

Literature Survey

SI NO.	Author(s) & Year	Dataset / Application	Paper Title	Method / Technique Used	Limitations	Key Findings
1	Cagliero et al. (2021)	Learning analytics/course data	Predicting Student Academic Performance by Means of Associative Classification	• Associative classification • Early exam-success prediction	• Course/exam focused • No continuous Digital Twin or intervention history	• Early data can predict exam success • Supports targeted help for at-risk students
2	Karalar et al. (2021)	Moodle + Adobe Connect	Predicting Students at Risk of Academic Failure Using Ensemble Model During Pandemic in a Distance Learning System	• Ensemble ML • GB, QDA, DT, RF, ET, LR, ANN	• Pandemic/distance-learning context Limited • continuous multi-source monitoring	• Ensemble model identified at-risk students • Asynchronous activity was highly informative
3	Park & Yoo (2021)	University online-learning data	Early Dropout Prediction in Online Learning of University Using Machine Learning	• ML classification • Early dropout prediction	• Online-learning focused • No persistent student-state model or intervention loop	• ML can identify potential dropouts early
4	Dabhade et al. (2021)	Academic, personal and behavioral data	Educational Data Mining for Predicting Students' Academic Performance Using Machine Learning Algorithms	•Educational Data Mining •Comparative ML	•Conventional prediction approach • No Digital Twin or closed-loop intervention	• Multiple student factors support performance prediction

5	Tamada et al. (2022)	Predicting Students at Risk of Dropout in Technical Course Using LMS Logs	• Random Forest, DT, NB • K-means + feature engineering	• 394 students • 1 million Moodle logs	• RF performed strongly • LMS engagement was useful for early prediction	• LMS/technical-course focused • Broader academic and behavioural events not integrated
6	Yağcı (2022)	Educational Data Mining: Prediction of Students' Academic Performance Using Machine Learning Algorithms	• Educational Data Mining • Supervised ML	• Student academic dataset	• ML identified patterns linked to academic achievement	• Dataset-level prediction • No continuously updated student state
7	Oqaidi et al. (2022)	Towards a Students' Dropout Prediction Model in Higher Education Institutions Using Machine Learning Algorithms	• ML classification • Feature selection + evaluation	• Higher-education student data	Feature selection and model choice affect dropout prediction	• Prediction focused • No Digital Twin or intervention feedback
8	Segura et al. (2022)	Machine Learning Prediction of University Student Dropout: Does Preference Play a Key Role?	• SVM, DT, ANN, LR • Feature analysis	• University of Madrid • 28 variables	• First-semester performance improves dropout detection • Course preference is relevant	• Very-early prediction remains limited • Single institutional context

9	Katsuragi & Tanaka (2022)	Dropout Prediction by Interpretable Machine Learning Model Towards Preventing Student Dropout	• Interpretable ML • Learning+ communication features	• Online-school data • Daily learning + Slack activity	• Behaviour and communication can support dropout prediction	• Online context • No multi-source Digital Twin
10	Su et al. (2022)	Applying Machine Learning Tecnologies to explore Students Learning Features and performance Prediction	• RF, SVM, LR, NN •Correlation analysis	• Programming-course quizzes + system logs	• Learning behaviour correlated with performance • NN performed well	• Programming-course focused • Limited broader academic/behavioural data
11	Plak et al. (2022)	Early Warning Systems for More Effective Student Counselling in Higher Education	• LR, additive LR, SVM, RF • Early-warning + counseling	• VU Amsterdam • Two first-year cohorts	•Prediction can support student-specific counselling	• Institution-specific •Intervention depends on local counselling practice
12	Qin et al. (2023)	Early Warning of Student Performance With Integration of Subjective and Objective Elements	• Neural network • Multi-source data synthesis + cost-sensitive learning	• University records • Academic,library, registration, access-control data	• Combining academic + behavioural factors improved warning	• Institution-specific integration • Deep-model interpretability remains challenging

13	Cho et al. (2023)	A Study on Dropout Prediction for University Students Using Machine Learning	• Machine learning • Dropout prediction + feature analysis	•University student records	• ML identified useful dropout-related factors	• Dropout focused • No continuous state tracking or intervention loop
14	Balcioğlu & Artar (2023)	Predicting Academic Performance of Students with Machine Learning	• DT, SVM, NN • Ensemble model	• OULAD	• Ensemble achieved strong prediction performance	• Performance prediction focused • No persistent twin + XAI intervention workflow
15	Nagy & Molontay (2023)	Interpretable Dropout Prediction: Towards XAI-Based Personalized Intervention	• ML dropout prediction • Explainable AI	•Higher-education student data	• Interpretability can support personalized intervention	• Mainly prediction + explanation • No event-driven Digital Twin trajectory
16	Villar & Robledo Velini de Andrade (2024)	Supervised Machine Learning Algorithms for Predicting Student Dropout and Academic Success: A Comparative Study	• DT, SVM, RF, GB, XGBoost, CatBoost, LightGBM • SMOTE/ADASYN + SHAP	•Higher-education dataset	• Boosting models performed strongly • SHAP supported factor interpretation	• Single-dataset dependence • Minority-class prediction remains challenging

17	Vaarma & Li (2024)	Predicting Student Dropouts with Machine Learning: An Empirical Study in Finnish Higher Education	• ML prediction • Monthly longitudinal feature analysis	• Finish higher education •Transcript + demographic + Moodle data	• Credits, failed courses and Moodle activity were important • Feature importance changes over time	• Single-university confidential data • Requires continual updating
18	Tiukhova et al. (2024)	Explainable Learning Analytics: Assessing the Stability of Student Success Prediction Models by Means of Explainable AI	• Predictive learning analytics • SHAP + model-stability/drift analysis	• Student success prediction data	• SHAP helps assess feature importance and stability •Importance changes across contexts	• Explainability focused • Not a complete early-warning/intervention platform
19	Miranda et al. (2024)	Machine Learning's Model-Agnostic Interpretability on the Prediction of Students' Academic Performance in Video-Conference-Assisted Online Learning	• RF, SVM,GNB • SMOTE + SHAP	• 361 online-learning records	• RF performed strongly • SHAP identified important performance factors	• Small, context-specific dataset • Online/pandemic context limits generalization
20	Alalawi et al. (2024/2025)	An Extended Learning Analytics Framework Integrating Machine Learning and Pedagogical Approaches for Student Performance Prediction and Intervention	•Predictive ML + pedagogical intervention • Web-based learning analytics framework	• Course assessment data • Academic pilot	• Connects prediction with intervention • Web framework supports academics	• Intervention depends on academics • Continuous multi-source event integration remains open

Literature Summary

Major Methods

1. Many studies use static or institution-specific datasets.
2. Most focus mainly on prediction rather than continuous monitoring.
3. Limited integration of academic, attendance and behavioural events.
4. Few systems connect predictions directly with interventions.

Key Findings

1. ML can predict academic performance and dropout risk at an early stage.
2. Attendance, academic performance, LMS activity and behaviour are important indicators.
3. Combining multiple student data sources improves risk prediction.
4. XAI helps identify the factors influencing predictions.

Common Limitations

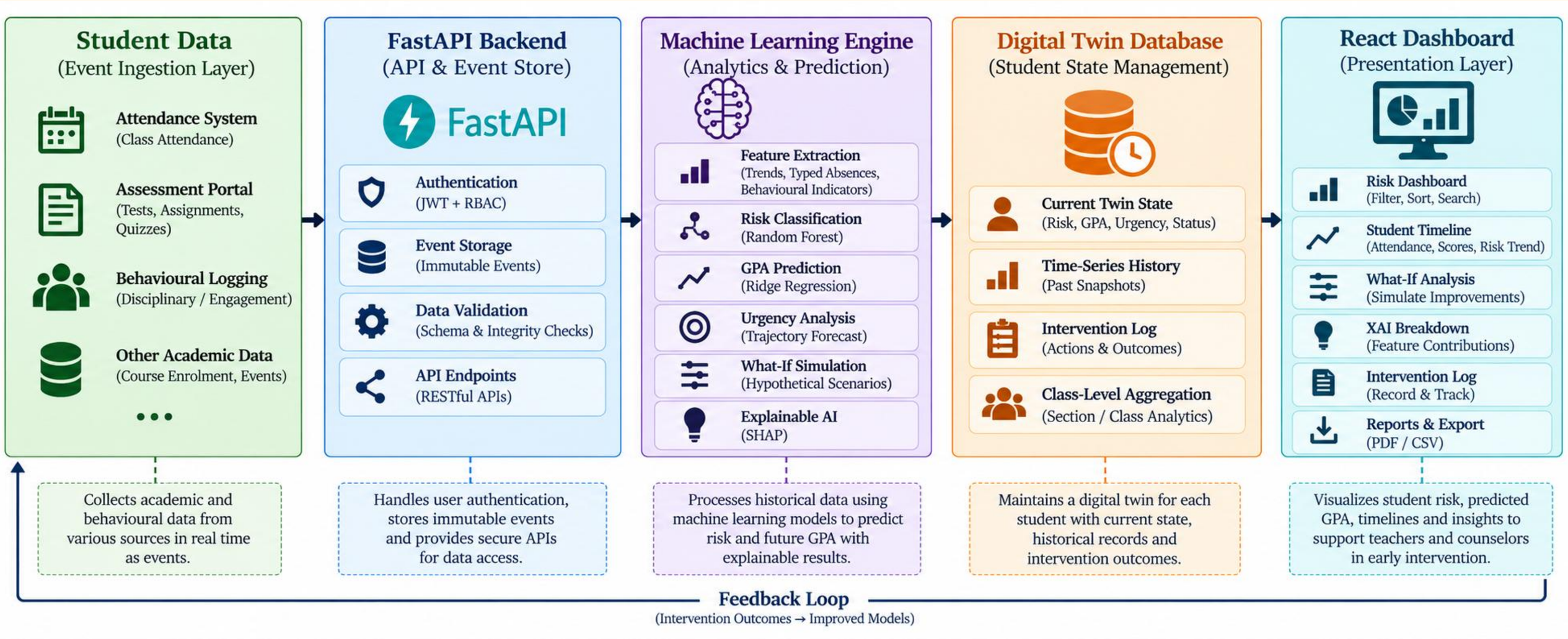
1. Machine Learning: Random Forest, Decision Tree, SVM, Logistic Regression
2. Ensemble & Boosting methods for dropout prediction
3. Educational Data Mining and Learning Analytics
4. SHAP / Explainable AI for prediction interpretation
5. Digital Twin & graph-based student pathway modelling

Relevance to Proposed Project

1. Our Digital Class Twin extends existing ML and Digital Twin approaches.
2. Uses event-driven continuous monitoring and time-series student state.
3. Provides Low/Medium/High risk, GPA prediction and trajectory-based urgency.
4. Uses SHAP, What-If simulation and intervention logging for actionable decisions.

Solution design

System Architecture



Workflow

Event Ingestion Layer

Attendance App • Assessment Portal • Behavioral Logging System

Immutable raw events

FastAPI Backend & Event Store

JWT Bearer Auth & RBAC • SQLAlchemy Event Store (immutable events table)

Event stream trigger

ML Engine & Analytics Pipeline

Feature Extractor → Risk Classifier → GPA Regressor → Confidence Calculator → XAI Engine

State snapshot write

Digital Twin Persistence Layer

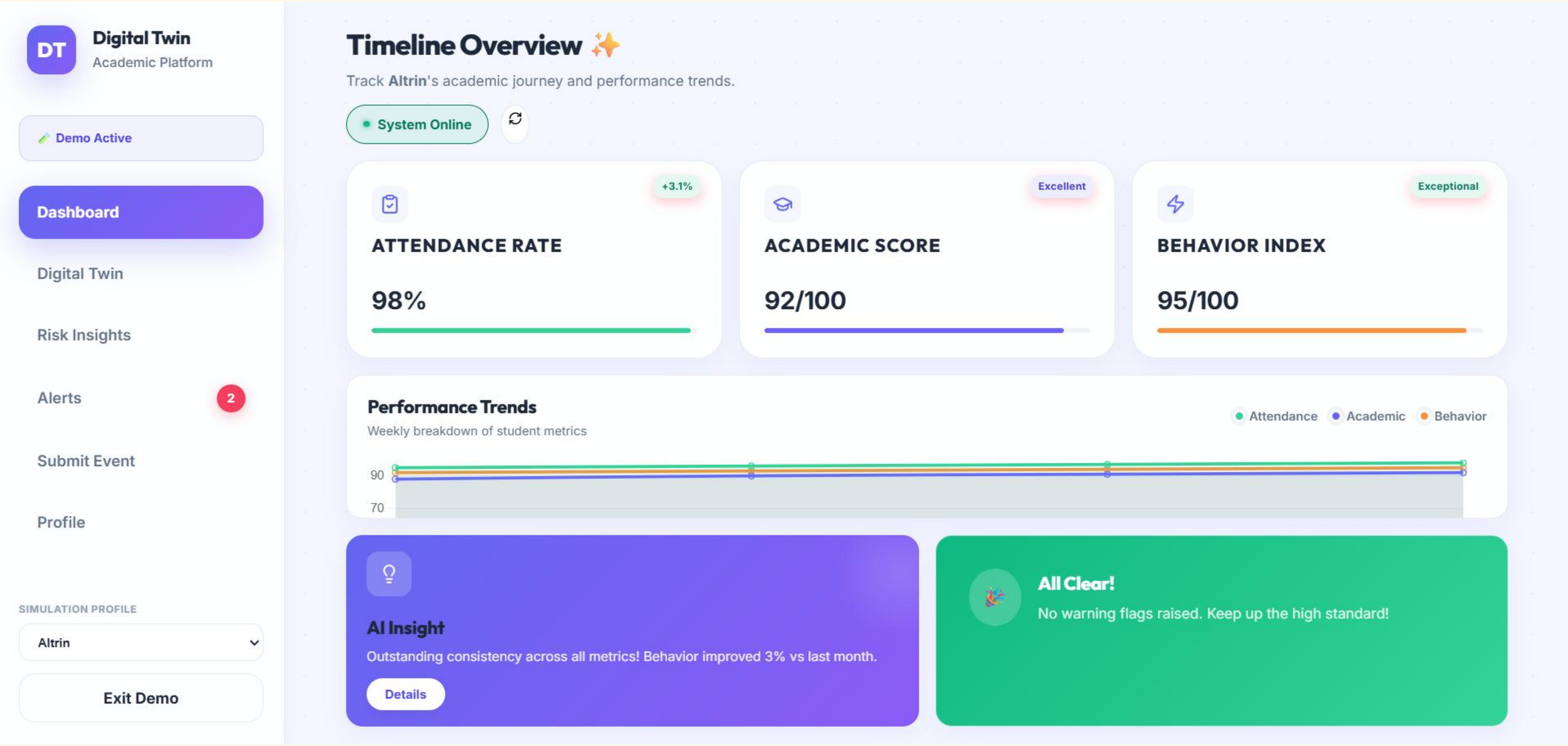
Current Twin State • Time-Series History • Closed-Loop Intervention Log

REST API / data read

Presentation Dashboard (React)

Risk Heatmap • Twin Timeline • XAI Breakdown • PDF Exporter

UI Design


DT
Digital Twin
Academic Platform
Demo Active
Dashboard
Digital Twin
Risk Insights
Alerts
2
Submit Event
Profile
SIMULATION PROFILE
Altrin
Exit Demo
Timeline Overview ✨
Track Altrin's academic journey and performance trends.
System Online
+3.1%
ATTENDANCE RATE
98%
Excellent
ACADEMIC SCORE
92/100
Exceptional
BEHAVIOR INDEX
95/100
Performance Trends
Weekly breakdown of student metrics
Attendance
Academic
Behavior
90
70
AI Insight
Outstanding consistency across all metrics! Behavior improved 3% vs last month.
Details
All Clear!
No warning flags raised. Keep up the high standard!

Implementation Details

Layer	Technologies Used	Key Packages / Specification
Front-End	React 18, Vite, Vanilla CSS, Recharts / Chart.js, Lucide Icons	Single-page application, component architecture
Backend Framework	Python 3.10+, FastAPI, Uvicorn	Async ASGI server, OpenAPI / Swagger docs
Database & ORM	SQLite / PostgreSQL, SQLAlchemy	Relational DB, time-series schema logging
ML Engine	scikit-learn, NumPy, Pandas, Joblib	Random Forest, Ridge Regression, softmax inference
Security & Privacy	PyJWT, Passlib (Bcrypt), OAuth2 Bearer Tokens	Role-based access control (Teacher / Counselor / Admin)

THANK YOU